Quantitative Finance
Day1: Course Overview, Introduction, Basics overview

Dr. Zahra Lakdawala

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Outline

1 Course Overview

2 Stocks - prices and returns

3 Python Labs

Course Overview

Do you belong here?

- Prerequisite/Co-requisite:
  - Probability
  - Differential Equations
  - Linear Algebra
  - Fundamentals of Computing
  - Fundamentals of Finance (not necessary)

Course Description

- You will learn to develop quant skills to implement financial theories.
- Apply sophisticated models in financial trading, pricing, risk and asset management functions

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References

- Paul Wilmott on Quantitative Finance, Wiley (2006)
- Reference Books: D. Higham,An Intrduction to Financial Option Valuation, Cambridge University Press (2004)
- F. Mc Neil, P. R & Embrects, Quantitative Risk Management, Princeton University Press 2005

Course Policy

- Assignments (take-home programming and analysis)
- Final project(take-home programming and analysis)
- Grading
  - Assignments: $(7 \times 10) = 70\%$
  - Final Project: 20%
  - Class Attendance: 10%

Course Topics

Modules

1. Review of Probability, DE, LA, Finance theory
2. Understanding Randomness
3. Stochastic Calculus
4. Understanding Markets, Derivatives, Options
5. Black Scholes, Binomial and CRR
6. Finite Difference Methods
7. Monte Carlo Simulations
8. Portfolio Management and Optimization
9. Capital Asset Pricing Model (CAPM), Fixed Income Securities, and Bond Pricing

From theory to practice

Python Labs

We will not only do this on paper. We will develop models and algorithms around it to understand the concepts fully. All assignments will contain programming parts.

Basic Financial Asset Classes

- Equities - Stocks
- Fixed Income, Bonds
- Commodities
- Currency
- Real Estate

Themes in Quantitative Finance

- Equity Derivatives - Risk due to Stock prices
- Fixed Income Products - Risk due to Interest Rates
- Credit Risk - Risk due to Credit Obligations Default
- Portfolio Analysis - How to manage risk and return
- Trading Strategies - How and when to trade assets

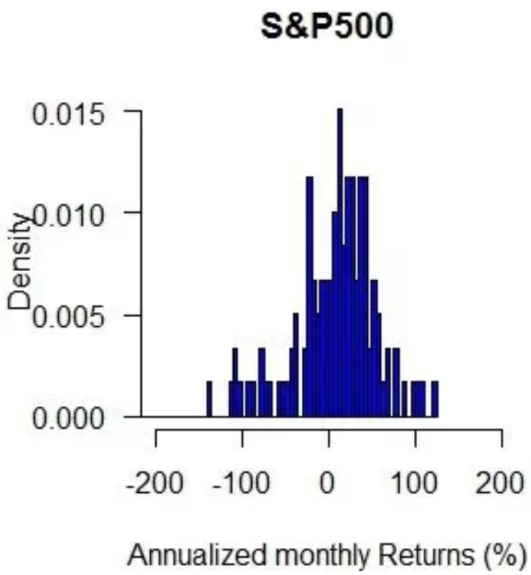
Stocks - prices and returns

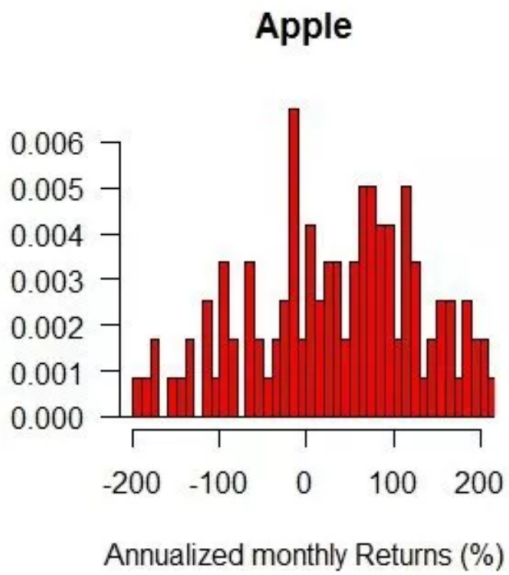
A Quick look at Stock Prices

- https://www.google.com/finance
- https://finance.yahoo.com/

Stock Returns

$$Return = \frac{P_t - P_{t-1}}{P_{t-1}} \tag{1}$$





Are returns normally distributed?

Python Labs

Python for first timers

- Install Anaconda: Follow instructions on *https://docs.anaconda.com/anaconda/install/*
- Launch Spyder
- Python is built upon libraries/packages
- Libraries can be installed using Anaconda prompt
  - pip install pandas-datareader
- Use of libraries in a python source file

Today's task

Stock prices

Get stock data from yahoo finance and plot it!